Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/31/2023
PI Name:	Paul, Anna-Lisa Ph.D.		
Project Title:	Epigenetic Adaptation to the Spaceflight Environment - Accumulated Genomic Change Induced by Generations in Space		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
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Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	10/26/2018	**End Date:**	10/25/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/25/2024 per NSSC information (Ed., 2/7/24) NOTE: End date changed to 10/25/2023 per NSSC information (Ed., 7/29/23) NOTE: End date changed to 10/25/2022 per NSSC information (Ed., 9/17/21)		
Key Personnel Changes/Previous PI:	Addition of Assistant Research Scientist Dr. Minqi Zhou to the project with expertise in epigenetic regulation in plants.		
COI Name (Institution):	Barbazuk, William Ph.D. (University of Florida, Gainesville) Ferl, Robert Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC19K0130		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Changes in the epigenome of plants, specifically DNA methylation patterns, are now known to occur in spaceflight. These changes are only beginning to be understood as far as their effect on physiological adaptation and productivity in spaceflight; however, the fact that DNA methylations are altered by spaceflight is clear, and disruption of methylation mechanisms affects growth in space. What is not yet understood, or even approached, is the trans-generational stability of these epigenetic changes, and whether establishing spaceflight methylation patterns within seeds can improve next-generation adaptability to spaceflight. Therefore, this proposal seeks to examine the transmission of DNA methylation patterns through seeds produced in plants grown on the International Space Station (ISS). Seeds produced on orbit will be used for second generation of spaceflight growth. This second ISS grow out, comprising seeds from spaceflight-grown plants together with seeds from terrestrially-grown plants, will determine whether second generation epigenetic changes continue to accumulate or stabilize. The second grow out will also reveal any spaceflight-associated changes in plant growth and development or gene regulation that are epigenetically regulated between plants grown from seeds produced on the ISS and seeds produced on the Earth. This experiment will determine whether growing a generation of plants in space confers an adaptive advantage to the next generation, similar to epigenetic adaptive advantage known to occur with some terrestrial plant stresses. This project is one of both discovery on a genome-wide scale and practical application to exploration needs. We propose unbiased whole genome bisulfite sequencing in Arabidopsis leaves across spaceflight generations to discover epigenetic modification patterns over generations, coupled with RNASeq examination of the comparable transcriptomes to assess spaceflight-associated DNA methylation on transcript profiles across generations. This will greatly advance fundamental understanding of the role of epigenetics in spaceflight adaptation. The practical applications arise from the fact that epigenetic imprinting occurs in plants exposed to environmental stress. Acquired DNA methylation patterns can transfer to the next generation, where they contribute to the ability of the progeny to cope with the stress experienced by the parental generation. Should that occur in spaceflight, we will have identified genetic elements that increase the adaptability of plants to spaceflight -- the first steps in developing spaceflight cultivars bred to better serve the roles of plants in life support. The tactical objectives of this proposal are: 1) Bisulfite whole genome sequencing to map spaceflight-induced changes in 5mCyt at single base-pair resolution for leaves of mature plants, to enrich the methylome data that we currently have for seedling leaves and roots. 2) RNASeq to assess transcriptome in the same samples, as differential methylation is linked to changes in gene expression that characterize the functional impacts of spaceflight across the entire genome in maturing leaves. 3) Collecting seeds produced on orbit and growing a second generation on orbit. The second flight grow out will be used to gather growth, development, gene expression, and epigenetic data on second generation plants on orbit. During both growth cycles on orbit, images of plant growth will monitor development. Elucidating multigenerational spaceflight methylomes contributes to the fundamental understanding of how biology responds and adapts to the spaceflight environment, and provides insight in the adaptive strategies plants use in the face of novel stimuli outside of their evolutionary experience. The long-term goal of this line of inquiry is to understand how the methylome contributes to the adaptive strategies of plants to spaceflight in order to breed plants that are specifically and better adapted to spaceflight in microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research contributes to a greater understanding of the role of epigenetic changes to the genome in response to novel or stressful environments. Further, it will be the first experiment to test whether the epigenomic changes observed in plants grown in a spaceflight environment can be inherited to the next generation, and confer an adaptive advantage to spaceflight physiology.
Task Progress:	Flight Experiment The flight experiment calls for two (2) separate launches and growouts of the Arabidopsis plants. These two were referred to during the operations at PH-03A and PH-03B. The Epigenetic Adaptation to the Spaceflight Environment - Accumulated Genomic Change Induced by Generations in Space (Epigenetic Adaptation to Spaceflight) launched its first deployment to the International Space Station (ISS) on NG-18, November 7, 2022 from the NASA Wallops Flight Facility (Wallops). The science carriers were stowed in room temperature storage and were unaffected by off-nominal transport that occurred during the flight in the cold stow sections. Once at the ISS, the science was transferred to a holding area until activation. Payload insertion into the NASA Advanced Plant Habitat (APH) hardware occurred on December 28, 2022, with the experimental profile starting December 29, 2022. Ground controls ran on a 6-day delay from flight. JEDIs Due to the protocol of the experiment, there were several crew interventions scheduled. The day before crew in-orbit operations, a daily overhead photo was tagged with the plants that were to be removed (thinning) or leaves to be harvested by the astronaut on the following day. These files, tagged as Java EDR (Experimental Data Record) Display Interfaces (JEDIs), are the starting point for in-orbit operations. Once operations begin, the crew discusses any changes to thinning or harvest selections with the Principal Investigator (PI) team over communications (COMs) at NASA Kennedy Space Center (KSC). PH-03A Experimental Procedures Plants were grown and harvested in the proposed bags in orbit. At 75 days, stems/bolts were cut, and the bottom of the bag closed. Seeds/stems in the drying bags were then left in the APH hardware with reduced relative humidity (RH) to allow the seeds to fully dry before transferring them to 4 deg storage. Once returned, seeds were de-stowed and returned to the PI laboratory at the University of Florida. Seeds were then cleaned from chaff and tested for germination rates. Seed Prep for PH-03B The individual quadrants were then individually sterilized. The dry sterilization process was utilized, which includes a 70% Ethanol wash for 10 minutes and then allowed to dry in a laminar flow hood for 2 hours. Post-sterilization tubes were then further divided into prepped Quadrants (Q) for planting. Ex. Q1:Flight, Q1:Ground, Q1:Backup, Q1:UF, and Q1:Test. This was repeated for each of the quadrants. The Qx:Test tubes had water added 24 hours after sterilization; we allowed to stratify for 24 hours and then planted to test for germination and sterility.

	Current Work Currently PH-03B is underway. This flight is due to be completed and returned to Earth Q4 of 2023. Once samples are returned, they will be processed according to the original aims of the project. Seeds will be tested, and leaf extractions (RNA and DNA) will be processed for sequencing.
Bibliography Type:	Description: (Last Updated: 11/20/2025)
Articles in Other Journals or Periodicals	Zhou M, Riva A, Gauthier MP, Kladde MP, Ferl RJ, Paul AL. "Single-molecule long-read methylation profiling reveals regional DNA methylation regulated by Elongator Complex Subunit 2 in Arabidopsis roots experiencing spaceflight." bioRxiv preprint server. Posted November 10, 2022. https://doi.org/10.1101/2022.11.09.515199 , Nov-2022